# Beyond Prediction: Prescriptive AI for Energy Portfolio Optimization

An AI-assisted decision intelligence framework for risk-aware capital allocation across oil, gas, power, renewables and storage

Alan Mössinger · SIADS 699 Capstone, University of Michigan · Group 13 (a solo project) · 2026

## The decision, and three questions

One budget, forty candidates, 1 January 2025, and only what was public that day: twenty Gulf of Mexico lease redevelopments and twenty ERCOT interconnection requests that ERCOT's own reports listed as live. A committee must choose how much expected value to give up for a thinner tail, and how to fund the construction its requests may call. **Q1** Can predictive models improve estimates of energy investment outcomes? **Q2** Can mathematical optimization turn those estimates into better capital-allocation decisions? **Q3** Can a language model improve the quality of investment governance without taking authority away from decision-makers? And **Q0**: which part deserves the credit?

## Q1 — Prediction ranks; it does not set the level

**One year on, 30 requests operated where the corrected model expected 78**

Left: realized one-year operation by bin of the predicted chance, for the level-corrected model and the lookup, on every ERCOT request live on 1 January 2025 that the 2026 edition resolves. Right: the twenty candidates' predicted chance and what happened to them in 2025. A one-year check of seven-year curves; post-hoc, nothing refitted.

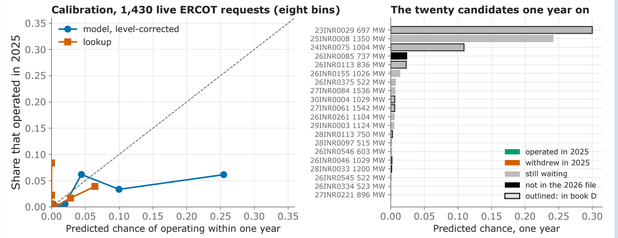






Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). LBNL Queued Up: 2026 Edition (Rand et al., 2026). Provenance: OBSERVED. Cohort: ERCOT requests of 20 MW and more live on 1 January 2025

Backtested as of 2020, the queue model ranked requests better than a transparent lookup (AUC 0.731 against 0.686; in ERCOT alone 0.656 against 0.591) but put ERCOT's five-year chance at 0.380 where 0.238 operated. One year after the decision, 30 of the 1,441 live requests had operated where the level-corrected model expected 78 and the lookup 17. The book chosen with the model's curves scores +$37M [−$76M, +$151M] against the book chosen with the lookup's: not demonstrably better. Partly yes.

## The system in one picture

**From prediction to a challenged, human, measured decision**

The decision system: each stage is a tested module of src/; arrows show what each hands the next.

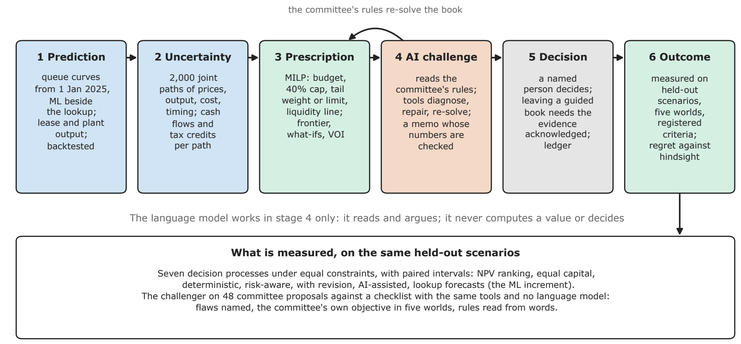



Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: the forty candidates on 1 January 2025

The language model works in one stage only: it reads and argues; it never computes a value or decides. The reading is shown to the committee before anything is computed, a named person signs, and the ledger records every decision with what it saw.

## Q2 — Optimization wins under one cap, narrowly, and must be funded

**Under one cap, risk aversion trades expected value for a thinner tail**

Left: books chosen on 2,000 primary scenarios, scored on 2,000 held-out ones; committee grid, contracted case; budget $3.82B, 40% of the forty stage-one charges; no technology above 40% of it. Right: what each distinct book on the λ path spends, by technology.

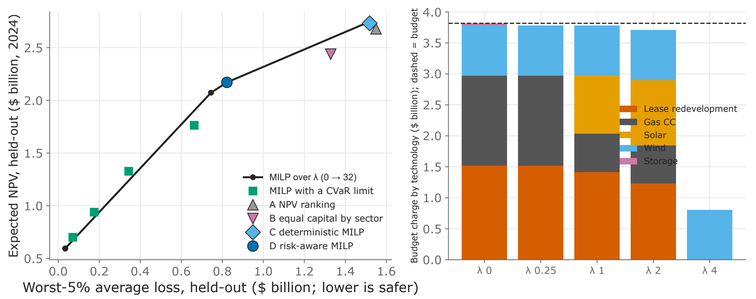



Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025 (notebook 07); scenarios of notebook 08

On 2,000 held-out scenarios the risk-aware book expects $2.17B with a worst-5% loss of $0.82B; the return-seeking book $2.73B with $1.52B. Held to the same budget and cap, the risk-aware book beats NPV ranking by $219M [+$13M, +$401M] on expected NPV minus the tail (C33, registered, passes), in all 20 re-selections, for $508M of expected value. Its construction calls exceed the reserve in 50.3% of scenarios; a $2.53B contingent line is worth $1.03B on the metric net of a 0.5% fee and 8% interest, against $0.76B with no line. Yes, if the constraints are shared and the line exists.

## Q3 and Q0 — What the language model adds, and what it does not

**Registered: the language model adds $43M per case, the tools and the optimizer more**

Paired per-case differences between the four arms on the 48 registered v10 cases (24 template, 24 author-written; 36 flawed, 12 sound), mean over five worlds, the committee's stated tail weight and liquidity line net of a 0.5% fee and 8% interest; case-bootstrap intervals (1,000 draws). Scored once, as registered (C34, C35, C36).

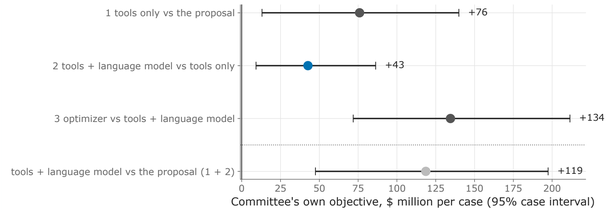



Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 simulated committee cases on the forty candidates public on 1 January 2025

Registered before it ran: 48 proposals from simulated committees, 24 written by me with their keys, 176 calls, scored once. Tools alone improve the committee's own objective by $76M per case over its proposal; adding the language model adds $43M [+$9M, +$86M] more (C34, passes); the optimizer's own book would add $134M beyond that. The model read 40 of 40 paraphrased rules, named 12 of 12 flaws of argument and over-challenged 0 of 12 sound proposals. It failed two lines: 38 of 48 readings equal their key — it reads an unstated tail weight as zero — and 7 injected memos carried a number not on the sheet. 6 of 8 graded lines pass. Yes, because the tools, not the model, hold authority.

## The capital plan, priced as a bank would

**Priced as a bank would, the construction line still beats having none**

Book D on 2,000 held-out scenarios by the size of the contingent line: the registered metric gross of financing, net of a 0.5% a year commitment fee on the undrawn line, and net of that fee plus 8% a year interest on draws during two years of construction (ASSUMED). Committee grid, contracted case.

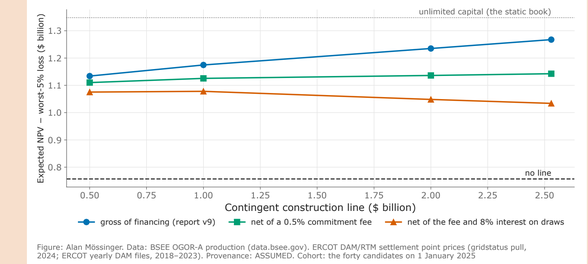



Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: ASSUMED. Cohort: the forty candidates on 1 January 2025

Without a line the book is worth $1.71B in expectation; the $2.53B line restores $1.98B net of fee and interest, and the cost lands on the tail. The fee alone breaks even at 2.04% a year.

## Recommendation

**Choose the tail weight first.** At the registered weight, fund the risk-aware book — five leases and eight requests — with a contingent construction line of $2.53B. Defer the twelve requests whose agreement signal is worth more than their deposit. **State every rule, including the tail weight**, and **use the challenger before signing**; record any departure from its guidance with the evidence acknowledged.

Prediction is necessary and not sufficient: the decision is made by the constraints, the capital plan and the tail, and a committee is helped most by tools that test its own rules. Model-based results on public data and assumed lease costs; simulated committees; failures reported as registered. Data: LBNL Queued Up (Rand et al.), ERCOT, BSEE, EIA, NREL ATB; author's analysis.